Majid Farhadi

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Education

University of Alberta, Computer Engineering — Edmonton, Canada, 2023 – 2026

Isfahan University of Technology, Computer Engineering — Isfahan, Iran, 2018 – 2023

Experience

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Graduate Research Assistant — Edmonton, Canada, 2023 – 2026, 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Structured Machine Learning Research Lab, Simon Fraser University, Graduate Research Assistant — Burnaby, Canada, 2023 – 2024, 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.

Embedded Systems and Networks Research Lab, Isfahan University of Technology, Undergraduate Research Assistant — Isfahan, Iran, 2021 – 2023, 2 years

- Designed an interface for a hardware accelerator for the time-slotted channel hopping (TSCH) protocol.
- Stacked on top of an IEEE 802.15.4 transceiver used in wireless sensor networks in IoT applications.
- Deployed on an Altera DE1-SoC board, featuring a Cyclone FPGA and a dual-core Cortex-A9 ARM.
- Integrated the ARM processors and FPGA fabric through a lightweight master-slave AXI bridge.
- Enabled full-duplex read/write communication between FPGA and ARM processor.
- Compiled C programs for the target ARM architecture using makefiles.
- Accessed the FPGA from C programs via virtual addresses provided by memory mappings.
- Offloaded computation from the processor and exploited the FPGA's high throughput.

Kian-Pardaz Naghsh-e-Jahan Engineering Company, Intern — Isfahan, Iran, 2021 – 2021, 1 year

- Designed a PCB and implemented a web client for a smart thermostat on an ESP8266-12E module.

Volunteer

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Sub-reviewer for Conference and Journal Papers — Edmonton, Canada, 2023 – 2026

- DDECS 2026
- ICLR 2026
- DATE 2026
- ASPDAC 2026
- IEEE TVLSI
- NeurIPS 2025
- MLCAD 2025
- ICCAD 2025
- IEEE TETC
- LAD 2025
- IEEE ESL
- DATE 2025
- ASPDAC 2025
- MLCAD 2024
- DFT 2024
- ICCAD 2024
- DDECS 2024
- ISCAS 2024
- VLSID 2024

Honors

Nobel Prize in Physics — Nov 1921

The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'

Royal Swedish Academy of Sciences

www.nobelprize.org/prizes/physics/1921/einstein/biographical